PRIVACY-PRESERVING SHOPPING INTELLIGENCE

How much recommendation quality survives when shopping data stays on the device?

Cloud vs federated vs on-device learning for Microsoft Edge — progress and decisions

Cloud Federated On-device

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THE QUESTION

Three ways Edge could learn from shopping behaviour

“How much shopping recommendation quality can be retained when moving from cloud-based to on-device or federated models?”

CLOUD

Events go to the server

One large model, trained and served centrally.

Best quality — no privacy.

FEDERATED

Events stay on the device

The device trains locally and sends only model updates.

Optional differential privacy.

ON-DEVICE

Nothing is uploaded

A small model is downloaded once and adapted locally.

Strongest privacy — least data.

Headline measure: retention = quality of a deployment ÷ quality of the cloud model.

Source: Research proposal §1; study description

THE JOURNEY

Four phases, each answering one question

1

Next category

Can we predict the next category?

Worked — but nothing published to compare with

2

Next product, cloud

How good can a cloud model get?

Strong models; compared with a published benchmark

3

Features

Which columns help?

Kept a small set that a device can compute

4

Deployment study

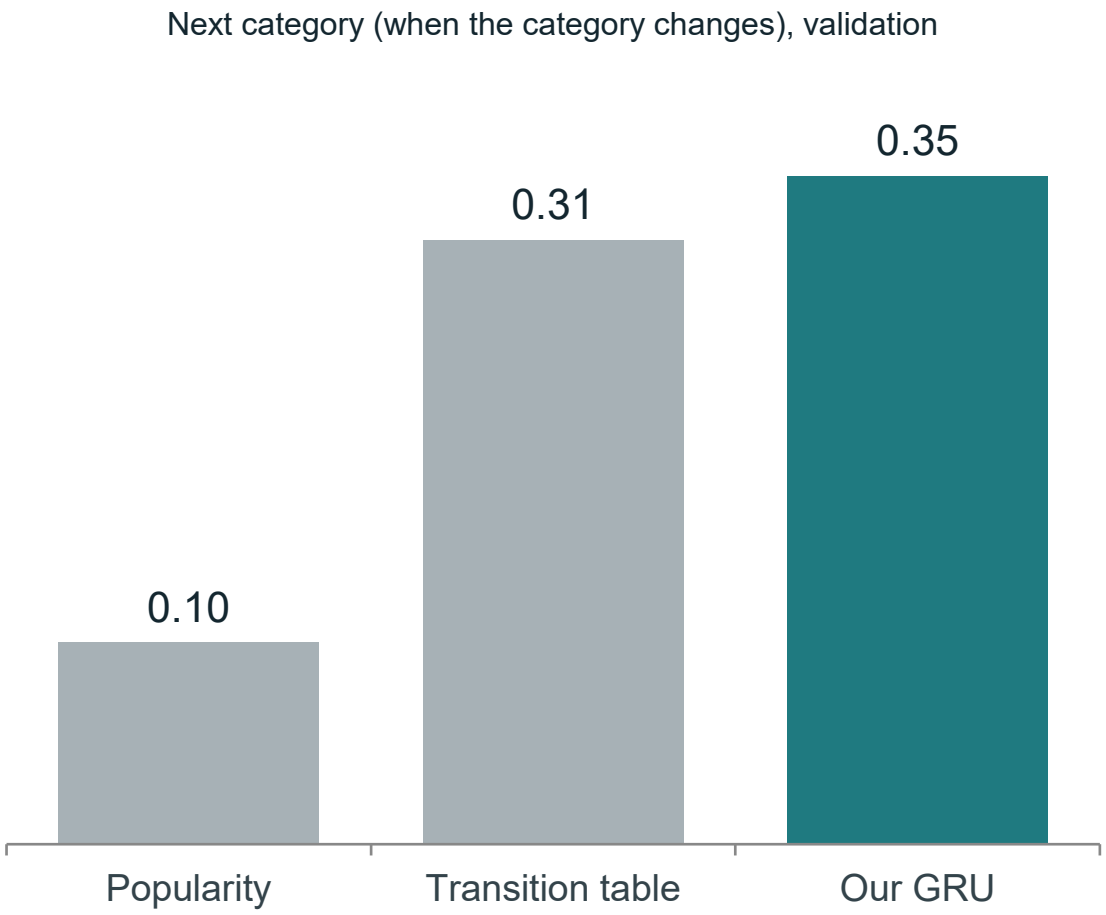
How much survives off the cloud?

Running now; main results sealed

Source: MVP report; central-model campaign reports; feature study; study description

EARLIER PHASE · October 2019 data

It worked, but we could not tell if it was good



Why we moved to next product

No published results

for next category on REES46 — no way to say if we are stronger or weaker

Too easy overall

most of the metric (≈ 82%) is solved by “repeat the same category”

A real benchmark exists

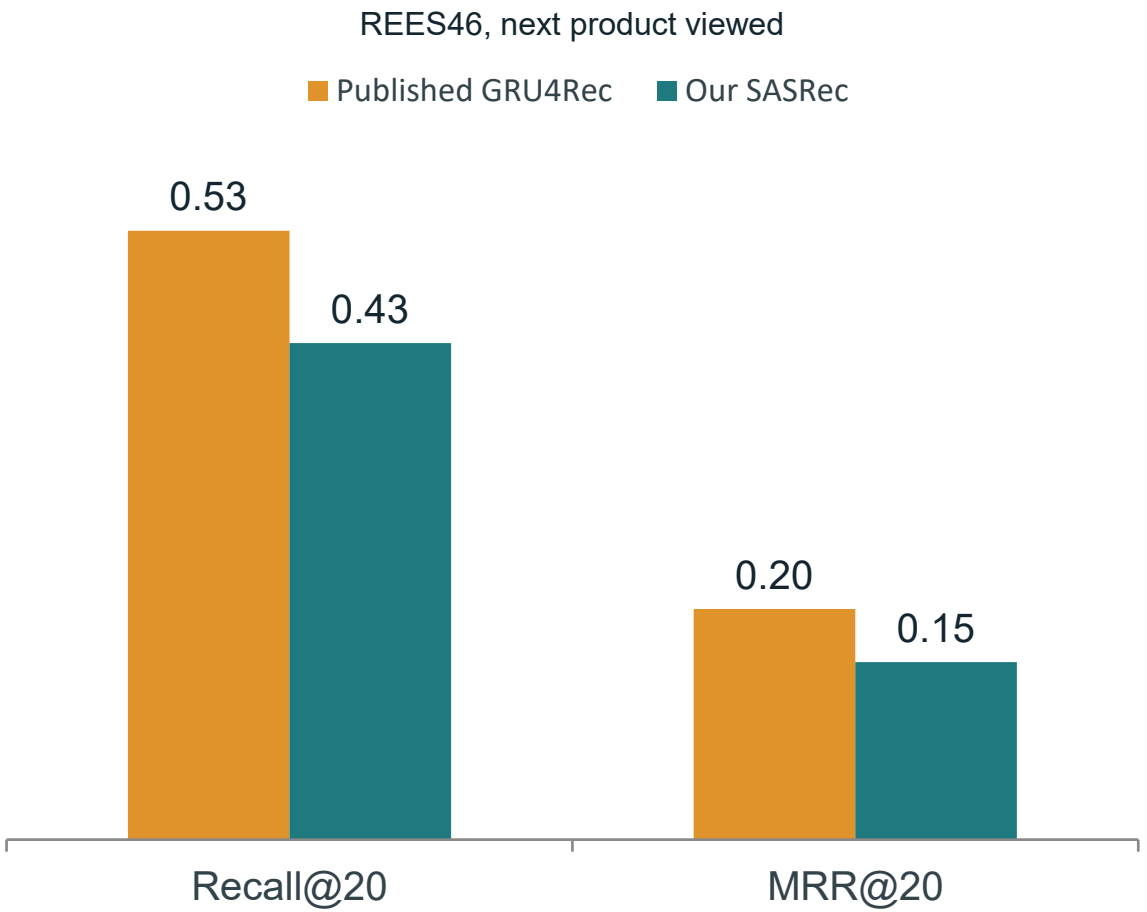
for next product on REES46 (GRU4Rec, RecSys 2023)

Agreed with the supervisor.

Source: GitHub repo: MVP report §2; evaluation-protocol decision (ADR-001); literature register

EARLIER PHASE · full data

Close to the paper, with a harder setup



Why ours is lower

Harder ranking	we rank the whole catalogue; new products always count as misses
Harder split	we predict the following week, not the last day
Less compute	a few epochs on two small GPUs — the paper is heavily tuned

Not a same-protocol re-run — now we have a 2× A40 server and have started experimenting.

Source: Hidasi & Czapp, RecSys 2023 (official GRU4Rec REES46 results) as recorded in the literature register; central-model report (validation week, per-event)

PHASE 3 — FEATURE ENGINEERING

From many columns to a small set a device can compute

Tokens	item · category · brand
Numbers	price · time since last view · session counts
User context	brand / category match share · price vs usual
Quality flags & masks	price missing · history available

58 → 24

input columns (earlier → now)

21 of 24

computable on the device itself

No calendar columns survived.

Removing all side features clearly hurts; adding more did not help — so we kept the set.

Source: Feature dictionary; feature study; feature search; independent review (features)

SCOPE

What changed from the proposal, and why

	Proposal	Now	Why
Task	Next category, purchase, offers	Next product viewed	Harder, standard, has a benchmark
Data	Five public datasets	REES46 (per-user, timestamped)	Each user can be simulated as one device
Metric	NDCG@10, HR@10, AUC	MRR@20 per user	Matches a browser serving one user
Federated	FedAvg, FedProx, FedBN, personalised	Same minus FedBN; + tuned optimizers	Our model uses layer norm, not batch norm
Privacy	“Various levels”	DP at ε = 8 and ε = 1	Makes privacy measurable

Kept: cloud upper bound · federated and on-device variants · cold start · communication cost · browser check.

Source: Research proposal §§3–4; study description; realism audit

PHASE 4 — DATA AND TASK

Next product, ranked over the whole catalogue

158,486

products ranked for every prediction

131,072

users in the main cohort — each one a simulated device

1.35 M

users behind the realistic full-pool cloud

Chronological split — the model never sees the future

Training — Oct to 15 Nov 2019	Validation — 16–22 Nov	Test — 23–29 Nov (sealed)

- Users with 5–100 events, so every simulated device has a realistic history.
- Tuning only on a separate calibration subset; the test week is opened once, after everything is locked.

Source: Data protocol and population lock; study description

METHOD

How we keep the comparison fair

1 Pre-registered

Rules and comparisons are written and sealed before the result is seen.

2 Same budget

Same data, features, evaluator and training exposure for every method.

3 Full catalogue

Exact ranking over every product; no sampled negatives.

4 Paired statistics

Same users for every comparison; two seeds for the main methods.

5 One test look

Everything is locked, then the test week is opened once.

6 Independent review

Every stage is checked by an independent reviewer before use.

Source: Evaluation contract; study description

DISCUSSION / DECISION 1

Task, benchmark and yardstick

1

Do you agree next product is the right task?
Next category had nothing published to compare with.

2

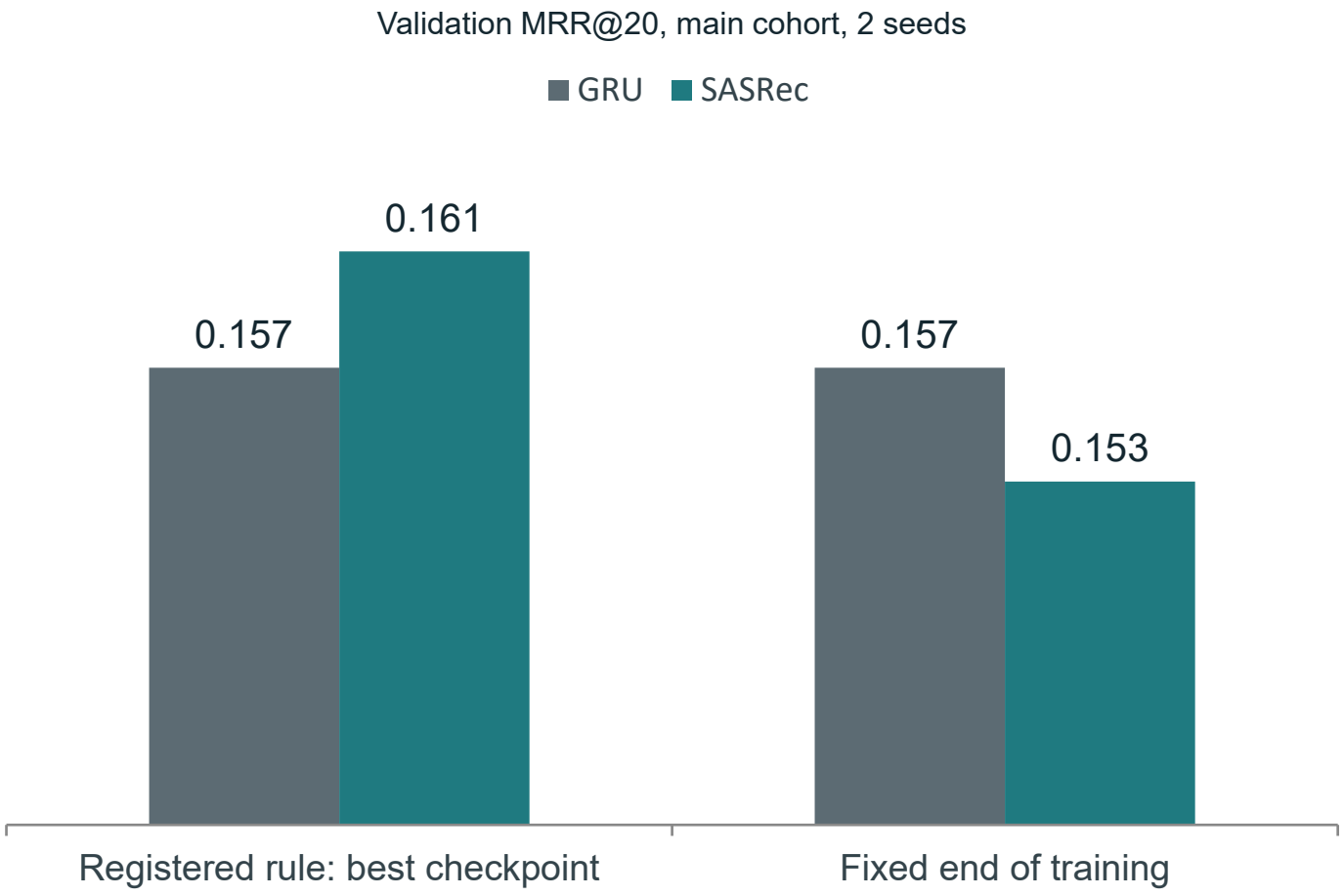
Should we re-run the published benchmark protocol on our new server?
It would tell us directly whether we are stronger or weaker.

3

Retention against which cloud: the matched 131k users or the 1.35M full pool?
We plan to report both — which should lead?

BACKBONE CONTEST

SASRec won the pre-registered contest against GRU



SASRec selected

by the rule fixed before any result; stable across both seeds.

Honest caveat

At the fixed end of training GRU is ahead — disclosed in the thesis.

Source: Contest result; contest diagnostics (decision log, 28 Sep)

MODEL SIZES

A large cloud model and a device model sized for a browser

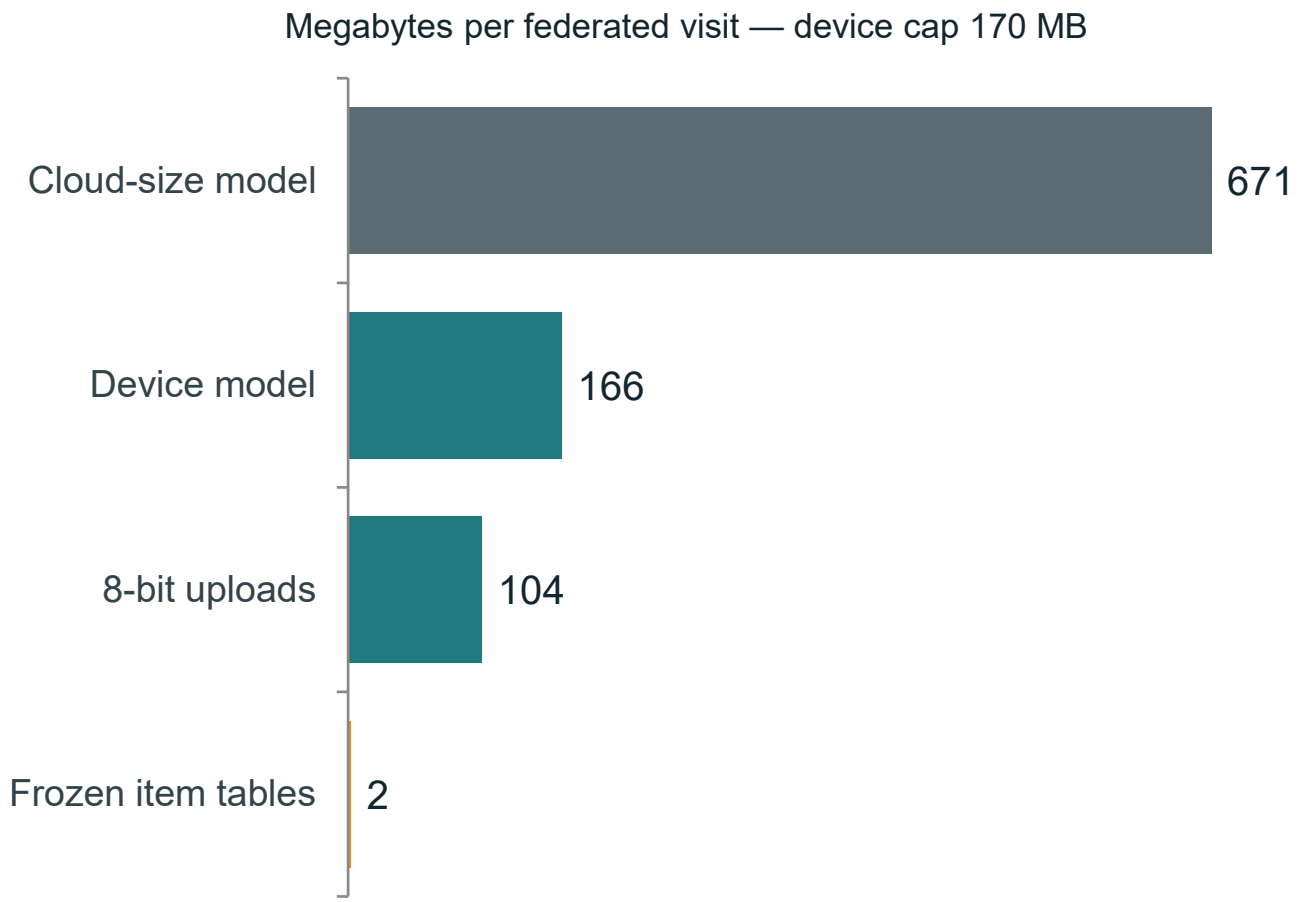
84 M

parameters — cloud model

21 M

parameters — device model

Only the device-size model fits the cap.



Source: Architecture lock (parameters, bytes per visit); review synthesis (frozen tables)

WHAT IS COMPARED

Every method, placed by what leaves the device

Privacy level	Methods	What leaves the device
Cloud	Large cloud model · small cloud model · full-pool cloud	Raw events
Cloud-trained, runs on device	Small model shipped to the device (ONNX)	Raw events, once
Pretrained on opted-in users	Pretrained → + federated → + frozen item tables	Opted-in: events · others: updates
Federated	FedAvg · FedProx · personal vector · 8-bit uploads	Model updates
Federated + DP	DP at ε = 8 and ε = 1	Noised updates
On-device	Local-only model · models fine-tuned on the device	Nothing
Baselines	Last viewed item · popularity · session kNN	—

Every method is also compared with "last viewed item" — a free device rule with no training.

Source: Privacy matrix (pre-registered, revised after the independent review)

DISCUSSION / DECISION 2

Which deployment should be the headline answer for Edge?

1

Plain federated, or pretrained-then-federated, or cloud model fine-tuned on the device?

The independent review recommends the pretrained branch — it mirrors how browsers ship features.

2

Should “last viewed item” be the floor every method must beat?

It needs no training and sends nothing.

Context: comparison matrix

FEDERATED LEARNING

Our first federated setup was wrong — and we fixed it

WHAT WE DID

- Started from an untrained model (random item tables)
- Reused the cloud's training settings
- No server optimizer; client optimizer reset every visit

WHAT IT CAUSED

- Federated far below the cloud on calibration users (≈ 20%)
- Even below the free “last viewed item” rule
- A warning from tuning users — not a final result

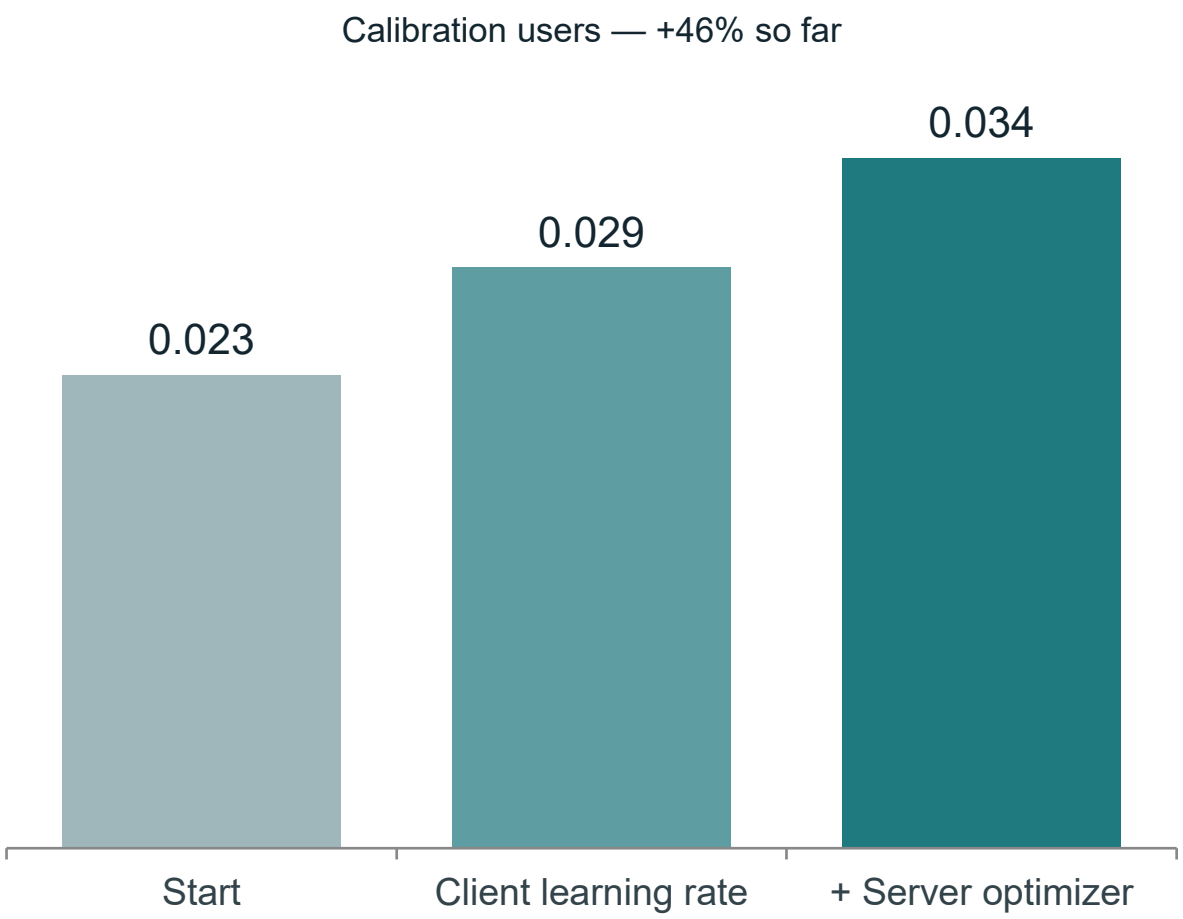
WHAT WE CHANGED

- Tuning ladder: +46% so far (learning rate, then server optimizer)
- Pretrained model ready — federated starts from it next
- Fixed before any final result is read

The gap mainly reflected our setup, not federated learning itself.

Source: Realism audit; federated ladder research; independent review; ladder register

One change at a time, each kept only if it helps



Stages

Client learning rate	done
Server optimizer	done
Learning-rate shape × client optimizer	running
Local passes and batch size	next
Start from the pretrained model	next
Freeze the item tables	next

A stage stays only if it gains ≥ 0.002.

Source: Improvement-ladder register; ladder stage registrations

PRIVACY

What the guarantee covers — and what it does not

ε = 8

moderate

ε = 1

strong

- Per-user guarantee: one user's whole history barely changes the model.
- Adversary: an honest-but-curious browser vendor.

Stated limits

- Without DP, an update can reveal which products a user viewed.
- Secure aggregation is not simulated.
- Frozen item tables reduce this leak, not remove it.

Source: Privacy statement and threat model (registered after the review)

REALISTIC DEPLOYMENT BRANCH

How a browser would actually ship it

1 Pretrain on opted-in users

≈ 1.2M users who share data; none of them in the main cohort

2 Federated on private users

Start from the pretrained model; private users send only updates

3 Freeze the item tables

Only a small encoder moves: about 1 MB each way

This directly fixes our first mistake: federated training no longer starts from an untrained model.

Judged against the full-pool cloud. Pretraining has finished; federated training from it is next. Results sealed until the first read.

Source: Warm-start registration and its review; pretraining cohort build

DISCUSSION / DECISION 3

Federated effort, privacy and the deployment story

1. **How much tuning effort is fair for federated learning?**
   Our ladder stops when a stage gains less than 0.002.

2. **Is DP at ε = 8 and ε = 1 enough, or do you want a curve?**
   One more level costs about half a day (estimate).

3. **Is pretraining on opted-in users a realistic assumption for Edge?**
   If yes, it becomes the headline Edge answer.

What runs on the device, and who it serves

On the device

Local-only model

a control, not a product

Cloud model, fine-tuned locally

download once, adapt

Pretrained model, fine-tuned locally

cleanest download-only option

Export — ONNX, full and 8-bit	done
Latency — single-thread CPU	running
Browser — Edge on a laptop	next

Cold start

Users who never took part

does the model still serve them?

Users with no history at all

what can any method do?

Short vs long histories

how many events until quality catches up?

A controlled test cuts histories to fixed lengths, so the answer is not just "heavy users differ".

Source: Study description; privacy matrix; cold-start registration

INDEPENDENT REVIEWS

7/10 today, 8–8.5 after the registered fixes

Strengths

- Pre-registration and one sealed test look
- Exact evaluation — no bug found
- No data leakage

Weaknesses found

- Federated setup too weak
- Privacy claims over-stated
- Device evidence from a server CPU

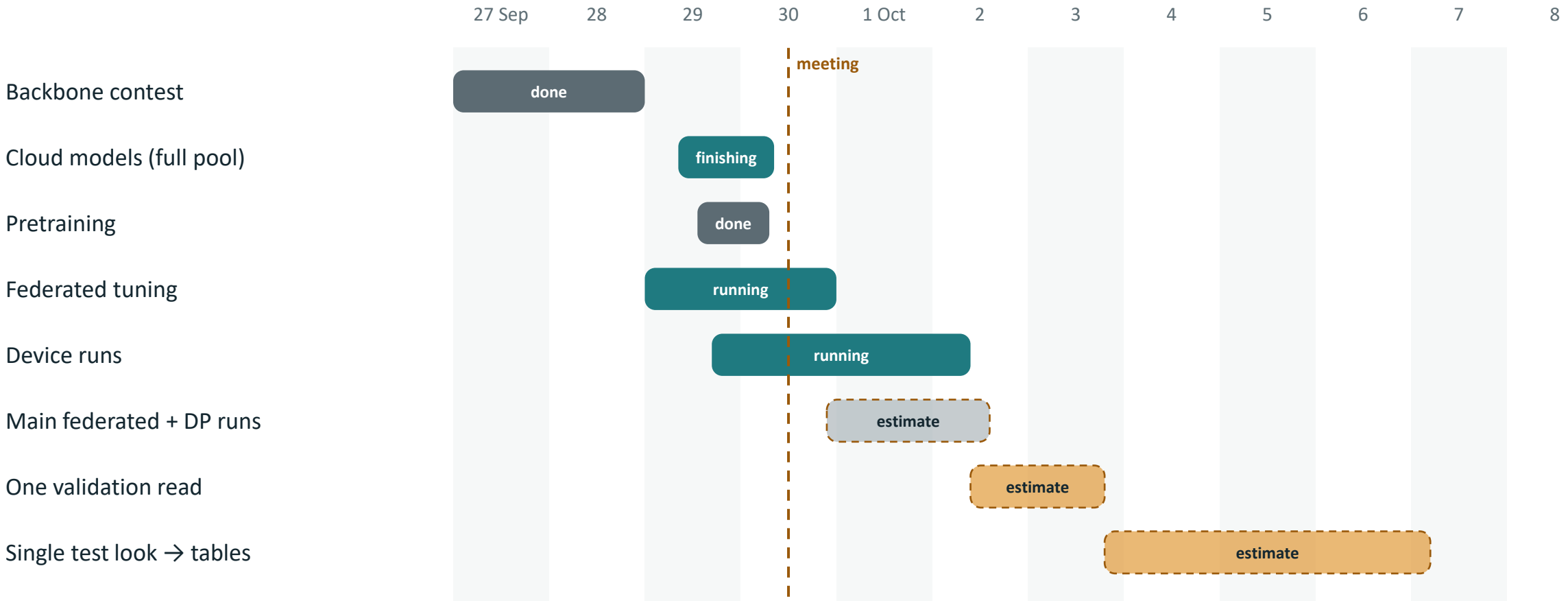
Already fixed or registered

- Tuning ladder and pretrained start
- Threat model and privacy matrix
- Realistic deployment branch promoted

Source: Multi-expert review synthesis (14 parts)

TIMELINE

What is running now and when results arrive



Future dates are estimates: validation read ≈ 2–3 Oct · test look ≈ 4–5 Oct · final tables ≈ 6–7 Oct.

Source: Master plan (measured run rates); decision log run status to 30 Sep 08:30 UTC

DISCUSSION / DECISION 4

Priorities if time is short

1

If we must cut, what should go first?
Never cut: federated, local-only and cloud fine-tune with two seeds, the lock and the single test look.

2

Is one browser run on a laptop enough device evidence?
Or should we add a second machine?

3

Can you pre-approve the Queen's AI-use disclosure?
Required front matter for the thesis.

Context: cut order is pre-registered

SUMMARY

Where the project stands

DONE

- Next category → next product, with reasons
- Compared with the published benchmark
- Features reduced to a device-friendly set
- Backbone chosen by a registered rule

RUNNING

- Federated tuning: +46% so far, after fixing our first setup
- Pretraining finished → federated from it next
- Full-pool cloud models finishing
- Device runs (local-only model running)

NEXT

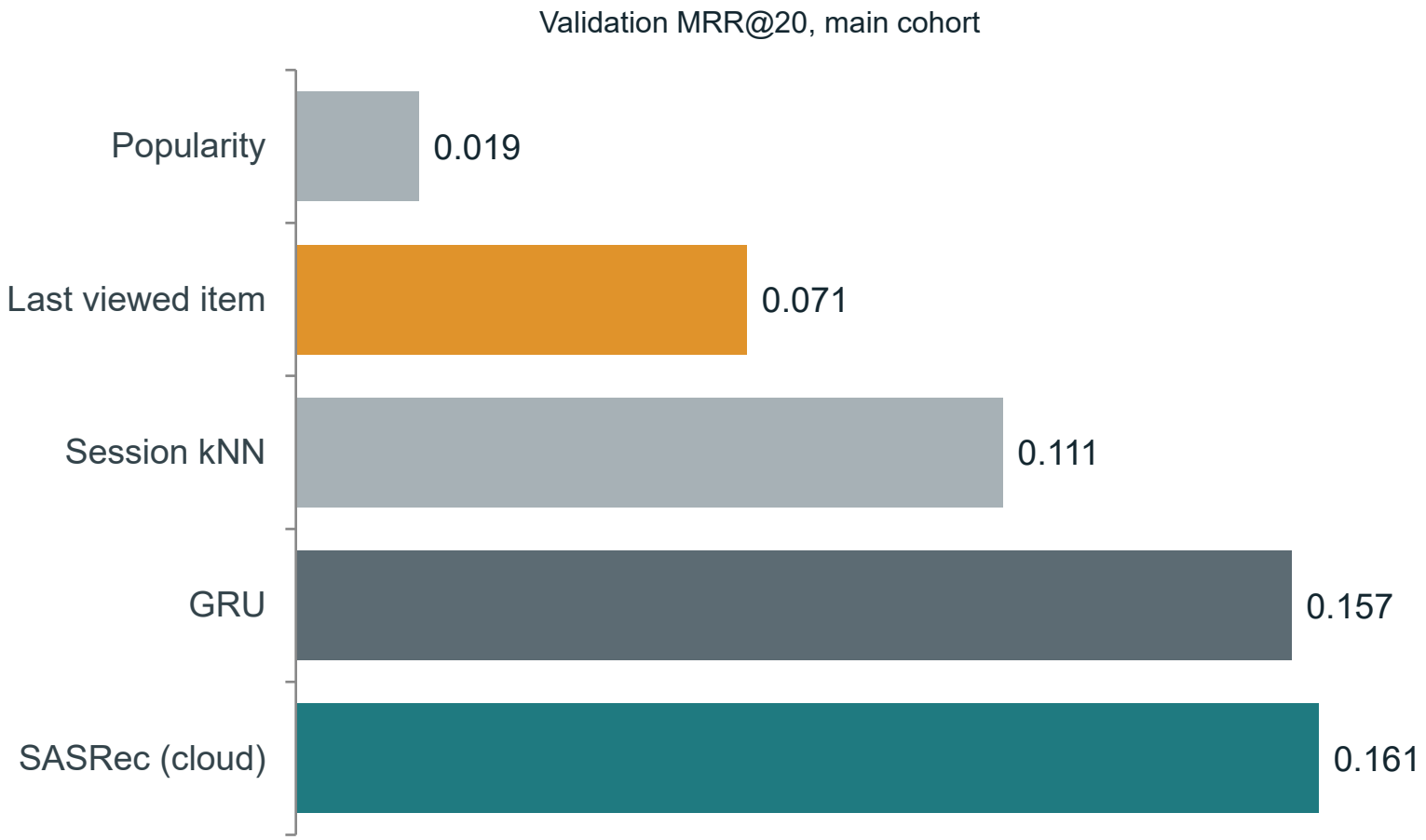
- One validation read (≈ 2–3 Oct, est.)
- Single test look (≈ 4–5 Oct, est.)
- Browser benchmark
- Thesis tables (≈ 6–7 Oct, est.)

Decisions today: task and benchmark · Edge headline · federated effort · DP levels · priorities

BACKUP

Details for likely questions

Simple rules vs the contest models (validation)

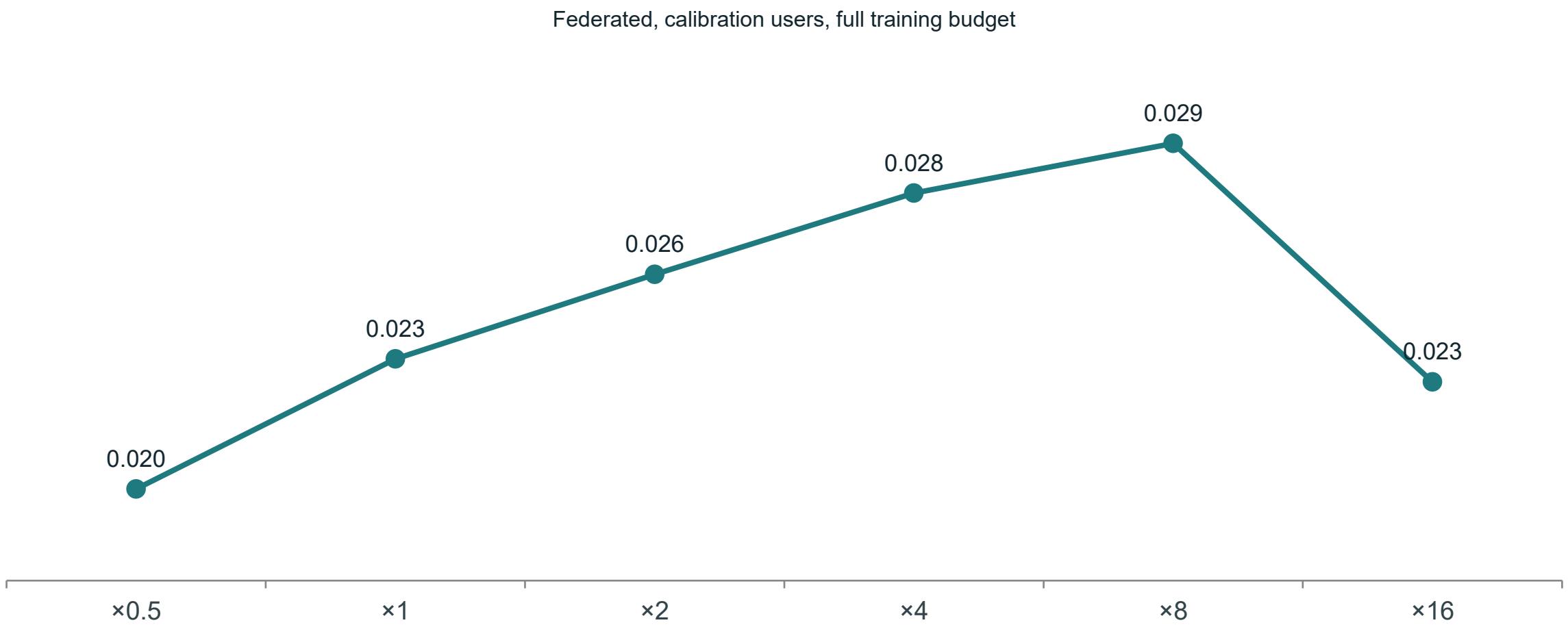


44%

of the cloud reached by “last viewed item” — for free

Source: Decision log (baselines and contest, 28 Sep)

Rise, peak at ×8, then fall



Source: Learning-rate selection files; ladder register